

Taelim Hwang

Software Engineer - AI Platform, Domain, Web Service

Seoul Korea, Republic of

(+82)10-3591-4626

ghkdxofla@gmail.com

[Github](#) | [Blog](#) | [Linkedin](#)

Profile

Software Engineer focused on Back-end for 7 years, now building AI platforms and content pipelines end to end. Interested in designing software architecture to increase efficiency. Knowledgeable in user interface, testing, and debugging processes. Proficient in various platforms, and languages especially on Python. Able to effectively self-manage during independent projects, as well as collaborate in a team setting.

Employment History

NextSecurities

September 2025 — Present

- Led end-to-end design and implementation of an AI content generation platform, from pipeline architecture to production
- Built and operated an in-house LLM Gateway serving multiple teams across 6 AWS accounts
- Architected AI video compliance review and evaluation systems, directing a 3-person team
- Built an on-premise 2-node DGX Spark GPU cluster and evaluated 9 open-source models across LLM, omni, TTS, STT, and video generation
- Ran a Databricks migration PoC and cut legacy infrastructure cost by ~\$26K/year

Lemonbase

October 2022 — May 2025

- Developed new features for the HR SaaS service platform
- Organized a development convention for Back-end chapter
- Solved security concerns for medium to large customers with SSO SAML integration
- Built a keyword-based search service w/ OpenSearch
- Designed and built a microservice with LLM(SaaS, OpenSource Model)
- Improved deployment experiences with CI/CD optimization

PortOne(Chai Corporation).

June 2021 — September 2022

- Developed a Legacy Payment Platform and programmed new features for connecting PG Providers
- Maintained New and Legacy Payment Platform
- Designed New Microservices for a New Version of Payment Platform with gRPC
- Pre-developed a New Simple Pay

LG Electronics. Inc.

January 2019 — June 2021

- Developed Information Display Management Web Solution(ConnectedCare)
- Researched and developed New IoT Platform Web Service
- Operated Data Management and analyzed IoT Data to create new businesses
- Pre-developed Mobile version of ConnectedCare(PoC)

ES Planit

June 2019 — December 2019

- Designed a core logic architecture for market price analysis platform using Koscom Open API
- Developed web platform for Stock market
- Migrated infrastructure AWS to Private Cloud Platform
- Operated infrastructure resource

Samsung Medical Center (Internship)

January 2018 — February 2018

- Shared tasks such as bioinformatics data preprocessing, and data analysis with R, Python, and Perl
- Developed Python automation tools for data collection

Education

Korea University, Seoul

March 2011 — February 2019

B.S. in **Computer Science** and Biotechnology

Skills

Programming Languages

Python, Golang, Rust, Kotlin, TypeScript, Java, PHP

Framework/Service

- Django(w/ Celery), FastAPI, LangChain, LangGraph, Spring, Node.ts
- OpenSearch, Kafka, gRPC, Github Actions

AI/Infra

- AWS Bedrock, LiteLLM, vLLM/Ray, DSPy, Langfuse
- Kubernetes(EKS), Terraform, ArgoCD, Databricks

Languages

Korean(Native)
English(Very good command)

Award-winnings

AWS Summit Seoul 2024 - Gen AI GameDay

May 2024

Won first place at genAI GameDay with **the highest score out of 33 teams**

ETH Seoul 2024 Hackathon

March 2024

Won an award for Mina Protocol w/ Prize **\$5,000**: Best overall Mina application or library built using Protokit framework in **TypeScript**

IoT Innovation Challenge

June 2018 — October 2018

won an Excellence Award for Chicken Fries Automation Process linked with cooking tools written in **Python**

SoC Robot War

March 2018 — June 2018

Advanced to the finals with programming a Real-time location tracking program written in **C++**

Project

AI Content Platform - NextSecurities

April 2026 — Present

Led architecture and implementation of an **AI-native content platform** converting real-time US equity market signals into short-form investor videos, from pipeline design through production.

- Designed a **9-stage generation pipeline** with an **18-state machine** and hard/soft quality gates, architected for **300-500 runs/day**.
- Architected **7 stage-specialized agents** (manuscript, script, scene-planning, script QC, video QC, persona extraction, character generation) on **LangGraph, DeepAgents, and the Claude Agent SDK**, served through **AWS Bedrock and Bedrock AgentCore Runtime** — each with its own system prompt, skill set, tool allowlist, and model parameters tuned per stage: raised the scene-planning agent's token ceiling to **32K** after 13-cut outputs measured 14K tokens and truncated at 16K.
- Designed and operated the platform's entire development infrastructure single-handedly — **EKS provisioned and managed as code with Terraform**, GitOps delivery via ArgoCD, and secret distribution through External Secrets.
- Built the platform's LLM observability stack by **self-hosting Langfuse** on EKS (web, worker, Valkey, and ClickHouse components with S3-backed event storage), giving the team end-to-end tracing and cost attribution across every agent and pipeline stage instead of relying on a managed vendor.
- Stabilized short-form video output from **desynced clips of around 10 seconds to reliable 60-second videos**, by tracing a silent frame-length fallback across the render boundary and re-deriving per-scene frame counts from **TTS audio duration** — a defect the renderer had reported as success, invisible to upstream quality gates.
- Authored and maintained a **20-skill plugin toolkit** that lets **non-engineers produce videos from a single natural-language request**, orchestrating the full signal → script → scene → render → QC → publish chain.
- Integrated the platform with **Databricks** so that signals generated from the financial knowledge ontology feed directly into video production, closing the loop from market data to published content.

AI Video Compliance Review System - NextSecurities

January 2026 — Present

Productionized and improved an **AI-based video compliance review engine** for financial content, evaluating text, audio, and video modalities against a regulatory ontology.

- Collaborated with the **AWS Generative AI Innovation Center** to benchmark multimodal foundation models on **416 human-labeled videos**, then restructured the rule taxonomy to reach **0.94 F1 (91.7% precision, 96.5% recall)** on flagged-video detection — up from 0.88 on the same set.
- Validated the production review model against a **598-video golden set**, achieving **96.2% precision** with a **2.7% false-positive rate** across 403 human-labeled cases.
- Integrated a domain expert's vibe-coded review dashboard with the evaluation server into a **single working system**, turning a standalone prototype into an operable review workflow for the compliance team.
- Refactored the inherited PoC engine into a production module with unit tests and unified LLM access through the internal gateway.
- Implemented a multimodal 3-stage pipeline (Twelvélabs Pegasus → Claude judge) over a **5-dimension, 25-rule compliance ontology**.

Self-Improving LLM Evaluation System - NextSecurities

February 2026 — April 2026

Architected and delivered a **self-improving evaluation system** for Korean financial video scripts, directing a 3-person team to a **59.8% error reduction** and shipping it to production.

- Architected the **3-layer evaluation design** (error analysis → LLM inference → gradient boosting calibrator) and the self-improvement loop that drove evaluation error **MAE from 0.838 to 0.337 (59.8% reduction)** across **5 iterations**.
- Directed **17 versions** of hybrid engine iteration, reducing MAE **22.1%** (0.728 → 0.567) and raising tolerance accuracy from **85.0% to 91.7%**; on the hardest criterion, MAE fell **47.6%**.
- Built and shipped the production evaluation service (**LLM-as-Judge, DSPy + LangChain hybrid**) with a **4-check compliance hard floor** and **12-dimension quality rubric** scored out of 60, hardening research-stage code into a deployed service.
- Set the technical direction and review bar for the team, catching security and error-handling defects in review; established the DSPy optimization pipeline that carried research results into production.

In-house LLM Gateway - NextSecurities

June 2026 — Present

Designed and operated a **LiteLLM-based LLM Gateway** unifying all internal Bedrock access with per-team isolation, budget control, and observability.

- Built a multi-account routing layer spanning **6 AWS accounts** with **10 base models** (Claude, GPT families), using cross-account AssumeRole with **zero static AWS keys**.
- Onboarded **2 teams — 5 engineers and 7 non-engineers** — with per-team model whitelists, virtual keys, and spend tracking backed by PostgreSQL and Langfuse tracing.
- Built and distributed a **macOS app** that lets engineers and non-engineers switch Claude Code between subscription and API modes in one step, removing the main adoption barrier for gateway-routed usage.
- Drove company usage past the qualification threshold of AWS's **6-week Bedrock for Claude Code promotion**, converting **\$9,000 of spend into \$18,000 in credits**.
- Diagnosed and resolved a production availability issue via profiling: reduced event-loop blocking from **79.9% to 2.4%**, pod restarts from **9 per 144 minutes to 0**, and liveness probe latency from **5.72s to 37-50ms**.
- Hardened the service with a full test suite, HA configuration (2 replicas, PDB, graceful drain), and a **5-scenario incident runbook**.

On-Premise GPU Cluster & Open-Source Model R&D - NextSecurities

January 2026 — Present

Built and operated the company's **on-premise GPU infrastructure** — a 2-node NVIDIA DGX Spark cluster — and ran open-source model evaluation across text, speech, and video modalities.

- Clustered **2 NVIDIA DGX Spark (GB10) servers** into a distributed vLLM/Ray deployment with **tensor parallelism across both nodes** over a dedicated **10Gbps** interconnect, scaling served context to **262K tokens** and tuning unified-memory allocation to eliminate OOM failures.
- Evaluated and served **9 open-source models across 5 modalities** on-premise: LLM (gpt-oss-120b, Qwen3.5-122B-A10B, Qwen3-Coder-Next, Gemma-4-31B), **omni/video understanding** (Qwen3-Omni-30B for video compliance review), **TTS** (Qwen3-TTS, 9 voices / 10 languages), **STT** (Qwen3-ASR), and **video generation** (LTX-Video-2) — the self-hosted Gemma-4-31B went on to **win a 500-article, 5-model English-Korean translation benchmark**.
- Verified that **FP8 quantization on Qwen3.5-122B-A10B** made a 122B-parameter model servable at production-grade quality within the cluster's memory envelope, extending what on-prem hardware could host.
- Connected the on-prem cluster to the **AWS R&D account over site-to-site VPN**, enabling n8n automation workflows to call self-hosted inference endpoints directly — replacing per-token vendor API spend with fixed-cost owned hardware for experimentation.
- Operated **7 internal services** (service portal, real-time GPU/VRAM dashboard, Open WebUI, vLLM Ray cluster, TTS API, ComfyUI, Ray dashboard) with access-control scripts, autostart recovery, and operational runbooks.

Internal AI Productivity Platform (Skills Hub) - NextSecurities

February 2026 — Present

Built and operated the company's **internal AI skills hub**, a shared marketplace distributing agent plugins and productivity tooling to engineering and non-engineering teams.

- Designed and maintained the hub registry serving **21 plugins across 5 teams** as primary maintainer.
- Curated a cross-domain toolkit covering code review, Jira/Lark workflow integration, language devtools, documentation, and security scanning — standardizing AI-assisted workflows company-wide rather than per-team.
- Led design of a company-wide **AI transformation pilot program** (5 ambassadors, 4-phase roadmap), including a constrained sandbox environment enabling non-engineers to build and ship their own automation tools.

Databricks ML Pipeline PoC & BI Dashboard - NextSecurities

October 2025 — December 2025

Planned and ran a **Databricks proof-of-concept** end to end, defining and operating **two evaluation scenarios** to decide whether the platform should adopt Databricks.

- Scenario 1 — **migrated the legacy platform's SageMaker ML pipeline to Databricks MLflow**, rebuilding the two-stage recommendation flow (**Two Tower retrieval + reranking**) with a Feature Store and MLflow experiment tracking and model registry, and validated it against the existing pipeline.
- Scenario 2 — **connected internal-network trading data to a BI dashboard** on Unity Catalog, proving the path from segregated transaction data to analyst-facing dashboards.
- Built a parallel ingestion pipeline across multiple financial data schemas, applying medallion (Bronze/Silver) layering with Delta Live Tables quality expectations.

Legacy Platform Handover & Cost Reduction - NextSecurities

September 2025 — March 2026

Led the handover of an external team's **9-system AI content platform** almost single-handedly, documented it from scratch, then decommissioned its infrastructure after migration.

- Reverse-engineered and documented **9 systems across 7 repositories** in **2 months**, producing **9 technical specifications** and **9 C4 architecture diagrams** — the sole reference for a platform handed over with no prior documentation.
- Took ownership of the platform's **Weaviate-backed RAG layer**, documenting how the agent stack ran semantic retrieval over embedded content and reproducing its vector search behavior well enough to migrate and then decommission it safely.
- Established the company's **handover management guideline** from this engagement, turning an ad-hoc takeover into a repeatable process for subsequent transitions.
- Decommissioned the legacy environment for a verified **~\$2,180/month (~\$26K/year)** reduction (**2 EKS clusters**, SageMaker endpoints and Canvas), executing triple-redundant backups of StarRocks metadata and Weaviate vectors beforehand with **zero data loss**.

Automated Code Review Bot TF - Lemonbase

April 2025 — May 2025

Developed and integrated an automated code review bot into the CI/CD pipeline to enhance code quality and enforce standards.

- Spearheaded a task force to build and deploy an automated code review bot operating within the **CI/CD process**.
- Designed the bot to automatically analyze **Pull Requests with LLM**, and post review comments directly.
- Ensured the bot's feedback aligned with and enforced the team's established **engineering review conventions by prompt engineering**.
- Contributed to a significant **reduction in post-deployment failures** by improving code quality before merging.

Iamchart, LLM-Powered Financial Analytics Server - ESPlanit(Side-Project)

March 2025 — Present

Developed a scalable **FastAPI-based server w/ LangChain, Bedrock** integrating multiple LLM models for automated financial chart analysis and technical indicator computation.

- Built a comprehensive financial data analysis system **supporting 12+ technical indicators (MACD, Bollinger Bands, RSI, Stochastics, etc.)** with automated chart pattern recognition.
- Architected multi-LLM integration supporting Claude 3.x, 4 series, Amazon Nova models, and Ollama with intelligent **rate limiting, smart routing and caching mechanisms**.
- Designed microservice architecture with domain-driven design principles, implementing separate modules for chart analysis, LLM API, and health monitoring.
- Built full-infrastructure and deployed microservices w/ CI/CD in an AWS.

Asynchronous Processing System Enhancement TF - Lemonbase

January 2025 — April 2025

Enhanced the asynchronous processing system using AWS MSK, boosting throughput via parallel processing.

- Redesigned the existing Kafka-based asynchronous processing pipeline, shifting from sequential to parallel execution.
- Architected the new system to allow a single consumer instance to process multiple messages.
- Achieved and validated up to an **8x increase in message throughput, reducing processing times by 80%**.

LLM Microservice Platform - Lemonbase

January 2024 — May 2025

Designed and built a **first microservices AI platform for LLM-based services**

- **Leading projects** and delivering successful results
- **Organized a hackathon** where everyone is free to develop services using the LLM
- Utilized **OpenSearch's Vector DB for RAG usage, LangChain and FastAPI**
- Advanced the service to work with both API call-based services provided by AWS such as **Bedrock and SageMaker** or by bringing up your own open source model
- Interacting with AWS Solutions Architects and earning credits toward LLM development and using those **credits (\$8,000)** toward the cost of running production infrastructure.

Analysis Report Service - Lemonbase

August 2024 — May 2025

Developed an **in-house first-time review analytics service**

- Build the service from the ground up, from **design to infrastructure**
- Designed DB dualization to handle heavy write loads
- Efficient organization of analysis logics through the application of asynchronous programming. Reduce **~20 minutes of synchronous configuration logic to ~5 minutes or less**
- Asynchronous job organization using **Kafka to efficiently handle multiple customers' analytics systems**

1:1 Service Improvement - Lemonbase

September 2023 — January 2024

Improving 1:1 service to increase the usage rate of customer members.

- Significantly improved usability with the addition of **Google Calendar recurring events**.
- Refactored codes to improve the internal API(**Django**).
- Built **the service's first keyword-based search system using OpenSearch** and **contributed to fix an issue in an open source library**

OpenAPI Improvement TF - Lemonbase

March 2023 — August 2023

Enable customers to integrate Lemonbase OpenAPIs(**Django**) to **quickly and conveniently integrate information** from external HRIS systems.

- Reduced logic that took over 30 seconds to add 1000 members to around **7 seconds for 3000 members (around 2 seconds for 1000 members)**.
- Systematized **error codes to minimize additional inquiries by engineers** in charge of integration with customers, and advanced the flow of OpenAPI request -> response check -> modification.
- Added **API Playbook** to provide guidance and **Dev Server** to provide a development environment for customers to use before integrating with production.

SSO SAML integration - Lemonbase

December 2022 — February 2023

Using SAML to connect customer's member information with the Lemonbase service(**Django**) for **easy login, high security, and higher service integration** rates.

- The client's administrators can **easily manage employees** by connecting to their existing SAML service accounts.
- Eliminated the adoption hurdle from a security perspective by enabling customers' security administrators to use the more secure security features offered by SAML services, such as **2-FA authentication**.

CI/CD Improvement TF - Lemonbase

November 2022 — January 2023

Improving **Github Actions CI/CD pipelines** to enable faster deployment and testing, and optimize pipelines for reliable deployments.

- Reduced deployment time from up to 20 minutes or more to **10 minutes or less**
- Optimized testing experience by skipping tests based on backend fixes, frontend fixes, etc.
- Speed up by organizing backend and frontend tests and builds **in parallel**
- Minimized cross-network costs by changing build image caching to AWS S3 storage, **saving up to \$5,000/month in costs**

Unified Payment System, Custom Pay - Chai Corporation

June 2022 — September 2022

Programming a simple server with socket(**Kotlin**) to communicate PGs and Card companies with dedicated lines.

- Became possible for the company to introduce a new payment method through the establishment of a dedicated line

Micro-service for New Payment Platform - Chai Corporation

September 2021 — June 2022

Refactored from legacy PHP-based services to new services based on microservice architecture, which fully designed a new **Transaction Gateway Service** and **Credential Management Service** for PG Provider. The time to link a new PG provider has been reduced from **3 months to 1 month**. Parallel work is possible through microservices, increasing productivity.

- Created an interface and architecture from scratch for interconnecting **microservices using gRPC** to reduce latency between other internal services
- Programmed a server with **Springboot based on Kotlin** from legacy PHP based services to make programming much easier and to be supported by various modules

Core System for Payment Platform(legacy) - Chai Corporation

July 2021 — November 2022

Added and maintained new features of the **CakePHP-based legacy server**. These features are currently used among the **top 3 major franchisees** in the company, with transaction volume occurring stably.

- Acquired domain knowledge about payment services, able to stably proceed with design for new microservices

IoT Platform Service - LG Electronics. Inc.

June 2020 — April 2021

Developed an IoT Platform Service with **Serverless Framework** for inter-connecting with IoT Devices, targeting structures similar to IFTTT. A lightweight server was quickly produced and configured for easy distribution, and the demonstration allowed it to proceed to **the commercialization stage**.

- Designed Service Core Engine through researching IPaaS service architecture, and developed an app for External Service Inter-connection(**Node.ts**) deployed to Azure Function and AWS Lambda with Serverless Framework

Data Pipeline for LG ConnectedCare - LG Electronics. Inc.

January 2020 — June 2020

Designed ETL Process of Raw Data generated by LG ConnectedCare solution including IoT device and network pattern, letting developers analyze imported data to derive meaningful insights. A task that previously took **more than 5 days has been reduced to 1 day**.

- Built an analysis environment using **Jupyter notebook Server**, **MongoDB** for raw data pre-processing with **Batch Scheduler(Python)**, and added a temperature management function through fan control from internal thermal data analysis

LG ConnectedCare Solution - LG Electronics. Inc.

March 2019 — June 2021

Developed new features for LG ConnectedCare Solution, increasing user convenience by detecting defects and controlling LED remotely. Preemptive after service is enabled through the solution without directly seeing defects that are difficult to check with the naked eye(at least **1 pixel error** can be detected).

- Front-end with **Vue.js** and Back-end API with **Python(Flask)**, modified **Stored Procedure for SQL Server**; maintained service on Azure Cloud, Azure Function, Azure PaaS Solutions and Blob Storage
- Developed a New Mobile Website(re-designed UI with **React.ts**) for CS Teams to control remote devices with remote controller UI, which received good reviews for a much more comfortable UX

Pantis, Financial Information Service Platform - ES Planit

June 2019 — December 2019

Designed a core logic and infra for **Market Price Analysis Platform** using Koscom Open API. It reliably collects and pre-processes **40 million stock data every day**, creating an environment that can be used directly for data analysis logic.

- Developed Stock Data Collector with Koscom Open API (**Python**), built pipelines with **SQL Server**, migrated from AWS to Koscom Cloud, and built CI/CD for scalable deployment

Extra-curricular Activities

Mentored developers at Himedia

March 2025 — Present

Provided technical mentoring for a project team of ~6 members, focusing on collaboration methods and AI development.

Code In Design

September 2023 — July 2024

Freelance developer at Code In Design, working on backend and infrastructure development. Participated in **5 projects** and successfully delivered.

hacktoberfest seoul 2023

October 2023

Participated in Hacktoberfest and **contribute to Rust-based open source**

Wanted OpenAPI Project Interview

September 2023

Participated in a developer interview for the **OpenAPI Project**

Depromeet(8th) at

June 2020 — December 2020

Developed and planned a Space recommendation service for work (**Back-end, Django**), released in Google Play Store and App Store. Built infrastructure with **AWS CDK** for a second side-project, Meditation Service.

KUCC(Korea University Computer Club)

September 2017 — February 2019

Machine learning exercise for NLP, data structures, and supported as a mentor for juniors.